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PERSONAL HYGIENE TOOL

Abstract

A personal hygiene tool comprises a handle extending along a first axis and comprising an opening formed at a first end, where a plurality of recesses are formed within the opening; and a tool head attachment at least partially retained within the opening and rotatable with respect to the handle around a second axis; wherein the tool head attachment comprises a latching mechanism comprising: a sliding latch slideable between an extended position where the sliding latch protrudes into one of the plurality of recesses and a retracted position where the sliding latch is disengaged from the recesses; a resiliently deformable element arranged to urge the sliding latch towards the extended position; and a first actuator configured to urge the sliding latch towards the retracted position against the resiliently deformable element.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present disclosure relates to personal hygiene tools and, in particular, to a personal hygiene tool with a rotatable tool head attachment.

BACKGROUND OF THE DISCLOSURE

[0002] Personal hygiene tools may encompass a wide variety of tools used for a large number of tasks in, for example, skin care, hair care, beauty and dental hygiene. It is conventional for such tools to have an elongated handle to be held by the user. Depending on the tool, it may be better for the tool head to be aligned with the longitudinal axis of the handle, set at a right angle to the longitudinal axis of the handle, or set at some other angle in between.

[0003] For example, a shaver will generally be arranged with the blade assembly aligned with the longitudinal axis of the handle, while a dermaplaner blade is usually set at a right angle to the longitudinal axis of the handle.

[0004] For some tools, such as a hair trimmer, it may be more comfortable and more stable to set the tool head at a different angle, depending on where the hair trimmer is used.

[0005] There is therefore a need for a personal hygiene tool with a rotatable tool head attachment.

SUMMARY OF THE DISCLOSURE

[0006] Features and advantages of the disclosure will be set forth in the description which follows, and in part will be obvious from the description, or can be learned by practice of the herein disclosed principles. The features and advantages of the disclosure can be realized and obtained by means of the instruments and combinations particularly pointed out in the appended claims.

[0007] In accordance with a first aspect of the present disclosure, there is provided a personal hygiene tool comprising: a handle extending along a first axis and comprising an opening formed at a first end, where a plurality of recesses are formed within the opening; and a tool head attachment at least partially retained within the opening and rotatable with respect to the handle around a second axis. The tool head attachment comprises a latching mechanism comprising: a sliding latch slideable between an extended position where the sliding latch protrudes into one of the plurality of recesses and a retracted position where the sliding latch is disengaged from the recesses; a resiliently deformable element arranged to urge the sliding latch towards the extended position; and a first actuator configured to urge the sliding latch towards the retracted position against the resiliently deformable element.

[0008] The first actuator may include a user-depressible button movable along the second axis.

[0009] The first actuator may include a camming surface set at an angle to a movement direction of the first actuator, and the camming surface is arranged to engage with the sliding latch and urge the sliding latch in a direction perpendicular to the movement direction of the first actuator.

[0010] The sliding latch may be an elongate element slideable along its length in a plane perpendicular the second axis.

[0011] An outer edge of each recess and a protruding end of the sliding latch may be bevelled or rounded.

[0012] The first actuator may be configured to urge the sliding latch to a point between the extended position and the retracted position where the sliding latch is partially engaged with one of the plurality of recesses.

[0013] The plurality of recesses and a protruding end of the sliding latch may have a circular form.

[0014] The personal hygiene tool may include a blocker which is engageable with the sliding latch in the extended position to prevent the sliding latch from moving towards the retracted position.

[0015] The personal hygiene tool may include a second actuator configured to move the blocker from an engaged position where the blocker is engaged with the sliding latch to a disengaged

position where the blocker is disengaged from the sliding latch.
[0016] The second actuator may include a user-depressible button movable along the second axis, where an activation direction of each actuator is along the second axis towards the other actuator.
[0017] The personal hygiene tool may include a second resiliently deformable element arranged between the first actuator and the second actuator to resist movement in the respective activation direction of each actuator.
[0018] The plurality of recesses may be arranged in an arc extending in a plane perpendicular the second axis.
[0019] The handle may include five recesses spaced equally apart.
[0020] The tool head attachment may defines a third axis extending between the second axis and a working point of the tool.
[0021] The sliding latch may be arranged to extend at 45 degrees to the third axis.
[0022] The second axis may be perpendicular to the first axis.
[0023] The tool may be a shaver, trimmer or scraper.
[0024] The tool head attachment may include a blade or blade assembly.
[0025] In accordance with a second aspect of the present disclosure, there is provided a personal hygiene system comprising a personal hygiene tool and at least one of a charging base formed to support the personal hygiene tool thereon and provide power to the personal hygiene tool; and a charging case formed to hold the personal hygiene tool inside and provide power to the personal hygiene tool. The personal hygiene tool comprises a handle extending along a first axis and comprising an opening formed at a first end, where a plurality of recesses are formed within the opening; and a tool head attachment at least partially retained within the opening and rotatable with respect to the handle around a second axis. The tool head attachment comprises a latching mechanism comprising: a sliding latch slideable between an extended position where the sliding latch protrudes into one of the plurality of recesses and a retracted position where the sliding latch is disengaged from the recesses; a resiliently deformable element arranged to urge the sliding latch towards the extended position; and a first actuator configured to urge the sliding latch towards the retracted position against the resiliently deformable element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0026] In order to describe the manner in which the above-recited and other advantages and features of the disclosure can be obtained, a more particular description of the principles briefly described above will be rendered by reference to specific embodiments thereof which are illustrated in the appended Figures. Understanding that these Figures depict only exemplary embodiments of the disclosure and are not therefore to be considered to be limiting of its scope, the principles herein are described and explained with additional specificity and detail through the use of the accompanying Figures.
[0027] Preferred embodiments of the present disclosure will be explained in further detail below by way of examples and with reference to the accompanying Figures, in which:—FIG. **1** is a perspective view of a personal hygiene tool according to an embodiment.
[0028] FIG. **2** is a perspective view of the personal hygiene tool.
[0029] FIG. **3** is a disassembled view of the personal hygiene tool.
[0030] FIG. **4** is a lateral section schematic of a first end of the personal hygiene tool.
[0031] FIG. **5** is a longitudinal section schematic of the first end of the personal hygiene tool according to a first embodiment.
[0032] FIG. **6** is a longitudinal section schematic of the first end of the personal hygiene tool according to a first embodiment.

[0033] FIG. **7** is a longitudinal section schematic of the first end of the personal hygiene tool according to a first embodiment.
[0034] FIG. **8** is a perspective view of a first actuator according to a second embodiment.
[0035] FIG. **9** is a longitudinal section schematic of the first end of the personal hygiene tool according to a second embodiment.
[0036] FIG. **10** is a perspective view of the second actuator according to a second embodiment.
[0037] FIG. **11** is a perspective view of the first end of the personal hygiene tool, according to a second embodiment.
[0038] FIG. **12** is a partially disassembled view of a tool head attachment of the personal hygiene tool according to a second embodiment.
[0039] FIG. **13** is an oblique section schematic of a tool head attachment of the personal hygiene tool according to a second embodiment.
[0040] FIG. **14** is a lateral section schematic of the first end of the personal hygiene tool.
[0041] FIG. **15** is a partially disassembled view of a blade holder of the personal hygiene tool.
[0042] FIG. **16** is a perspective view of the personal hygiene tool with a charging base.
[0043] FIG. **17** is a disassembled view of the charging base.
[0044] FIG. **18** is a plan view of the personal hygiene tool with a charging case.
[0045] FIG. **19** is a disassembled view of the charging case.
[0046] FIG. **20** is a perspective view of a kit including the personal hygiene tool, charging base and charging case.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0047] Various embodiments of the disclosure are discussed in detail below. While specific implementations are discussed, it should be understood that this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the scope of the disclosure.
[0048] FIG. **1** is a perspective view of a personal hygiene tool **1** according to an embodiment. FIG. **1** shows a personal hygiene tool **1** comprising a handle **100** and a tool head attachment **200**.
[0049] The handle **100** extends along a first axis. The first axis may be referred to as a longitudinal axis of the tool. The handle **100** may be generally cylindrical.
[0050] FIG. **2** is a perspective view of the personal hygiene tool **1**. As illustrated, the tool head attachment **200** is rotatable with respect to the handle **100** around a second axis. The second axis may be perpendicular to the first axis. The second axis may be referred to as a transverse axis of the tool.
[0051] The tool head attachment **200** may be configured to rotate through a maximum arc of 90 degrees. In some examples, the maximum rotation arc of the tool head attachment **200** may be smaller than 90 degrees, or may be larger than 90 degrees up to a maximum of around 180 degrees. The tool head attachment **200** may rotate in one direction only from a starting position in which the tool head attachment **200** is aligned with the first axis. The tool head attachment **200** may rotate symmetrically in either direction from the first axis.
[0052] The tool head attachment **200** may be configured to rotate to a plurality of fixed positions. The fixed positions may be equally spaced along the arc of rotation or may be distributed unevenly e.g. with an increasing or decreasing spacing between each position. As shown, the number of fixed positions may be five. In some examples, there may be a greater or lesser number of fixed positions.
[0053] The personal hygiene tool **1** may be a shaver, trimmer, scraper, dermaplaner or any other personal hygiene tool. In some examples, the personal hygiene tool **1** may be a dental hygiene tool e.g. a toothbrush or flosser, or a hair care tool e.g. a curling or straightening iron, or a beauty tool e.g. nail clipper, nail buffer, eyelash curler, eyebrow razor or threading tool. The tool head attachment **200** may include a blade or a blade assembly **10**. The blade assembly **10** may be a static blade assembly with one or more blades or a moving trimmer assembly. The blade assembly **10** or

other tool head may be fixed to the tool head attachment **200** or may be replaceable. The tool head attachment **200** may be configured to fit a plurality of different tool heads.

[0054] FIG. **3** is a disassembled view of the personal hygiene tool **1**. The handle **100** comprises an opening **101** formed at a first end. The tool head attachment **200** is at least partially retained within the opening **101**.

[0055] The handle **100** may be formed with a front handle portion **110**, a rear handle portion **120**, and an outer grip **130**. The front handle portion **110** may form the first end of the handle **100**. The front hand portion may be hollow. The first end of the front handle portion **110** may be open and may receive the tool head attachment **200**.

[0056] The tool head attachment **200** may include a first actuator **330**. The front handle portion **110** may include one or more additional openings **111** to access the first actuator **330**. In some examples, the tool head attachment **200** may be received and the first actuator **330** accessed through a single opening **101**. In the example shown, the tool head attachment **200** includes a blade holder with a single blade.

[0057] The rear handle portion **120** may form a second end of the handle **100**, opposite to the first end. The rear handle portion **120** may be hollow. The rear handle portion **120** may include an end cap **140**. The end cap **140** may be fixed in position or may be removable to access an interior of the handle **100** e.g. for changing one or more batteries.

[0058] The interior of the handle **100** may include one or more electronic components, for example, a power source, a charging interface, a circuit board, a controller, a motor, a user input, an indicator etc. The power source may include a fixed power pack, a DC input connection or a changeable battery. The charging interface may be a wired or wireless interface arranged at or near the end cap **140**, e.g. a wireless charging coil positioned within the end cap **140**. The motor may be connected with a clipper or shaver attachment and/or equipped with a vibration function. The user input may include one or more buttons to interact with the controller e.g. to activate the motor and or set a power level of the motor, e.g. corresponding to a level of vibration. The indicator may include one or more lights and/or displays and/or sound output to provide information for the user e.g. to indicate an active state, or a battery level of the tool.

[0059] The front handle portion **110** may be fixedly attached to the rear handle portion **120**. The outer grip **130** may be formed in one or more portions and arranged to fit around one or both of the front and rear handle portions **110**,**120**. Other suitable handle configurations may be contemplated by a personal skilled in the art.

[0060] FIG. **4** is a lateral section schematic of a first end of the personal hygiene tool **1**. The opening **101** may be formed with a curved inner wall. A cross-section profile of the curved inner wall may be formed as a section of a circle. A plurality of recesses **102** are formed within the opening **101**. The plurality of recesses **102** may be formed in the inner wall of the opening **101**.

[0061] The tool head attachment **200** comprises a latching mechanism **300**. The latching mechanism **300** comprises a sliding latch **310** and a resiliently deformable element **320**. The sliding latch **310** is slideable between an extended position and a retracted position. The sliding latch **310** may be an elongate element slideable along its length in a plane perpendicular the second axis.

[0062] The tool head attachment **200** may include an attachment frame **210**. The attachment frame **210** may be configured to support an attached tool head and support the latching mechanism **300** within the opening **101** of the housing. The attachment frame **210** may be generally circular. The attachment frame **210** may be positioned to abut with the curved inner wall to rotate within the opening **101**. In the extended position the sliding latch **310** may protrude beyond the circular extent of the attachment frame **210**. In the extended position the sliding latch **310** protrudes into one of the plurality of recesses **102**. In the retracted position the sliding latch **310** is disengaged from the recesses **102**.

[0063] The sliding latch **310** may have a forward end where the sliding latch **310** engages with the handle **100**, and a rear end where the sliding latch **310** engages with the resiliently deformable member. The resiliently deformable element **320** is arranged to urge the sliding latch **310** towards the extended position. The resiliently deformable member may be arranged at the rear end of the sliding latch **310**. The resiliently deformable member may be arranged between the sliding latch **310** and a compression surface. The compression surface may be formed by an inner wall of the tool head attachment **200**.

[0064] When the sliding latch **310** is in the extended position, the resiliently deformable element **320** may be fully or almost fully expanded. When the sliding latch **310** is in the retracted position, the resiliently deformable element **320** may be fully or almost fully compressed. The resiliently deformable element **320** may be a metallic or plastic spring e.g. a coil spring, wave spring or disc spring, or may be a deformable block formed of e.g. foam, gel, or deformable plastic material.

[0065] The latching mechanism **300** comprises a first actuator (not shown). The first actuator is configured to urge the sliding latch **310** towards the retracted position against the resiliently deformable element **320**.

[0066] In this way, the tool head attachment **200** of the personal hygiene tool **1** can be rotated between a number of fixed positions, which allows the tool to be used in a variety of different positions. In this way, the ergonomics and usability of the tools can be improved. A user can rotate the tool head attachment **200** into the position which is most comfortable and most stable for the desired task. For example, where the personal hygiene tool **1** is implemented as a shaver, the shaver head can be rotated between a normal longitudinal position to a right-angled position for improved stability in a fine detailing task. Where the tool head is replaceable, the tool head attachment may be rotated to the most suitable position for the task of the attached tool head, e.g. to a right angled position for a dermaplaning tool head.

[0067] FIG. **5** is a longitudinal section schematic of the first end of the personal hygiene tool **1**. FIG. shows the front handle portion **110**, the outer grip **130** of the handle **100** and the tool head attachment **200**. The tool head attachment **200** may include the attachment frame **210**, the sliding latch **310**, the resiliently deformable element **320** and the first actuator **330**. The attachment frame **210** may include an upper latch support **211** and a lower latch support **212**. The sliding latch **310** may be held in position between the upper latch support **211** and the lower latch support **212**.

[0068] The first actuator **330** may include a user-depressible button **331** movable along the second axis. The first actuator **330** may be supported by the upper latch support **211**. As shown, the user may apply a downwards force to move the button **331** and the first actuator **330** downwards along the second axis. FIG. **5** shows a first state where the button **331** of the first actuator **330** is not pressed.

[0069] The first actuator **330** may include a camming surface **330***a* set at an angle to a movement direction of the first actuator **330**. The camming surface **330***a* may be set at 45 degrees to the second axis. The camming surface **330***a* may be arranged to engage with the sliding latch **310**.

[0070] The sliding latch **310** may have a reaction surface **310***a* set at an angle to the second axis. The reaction surface **310***a* may be set at 45 degrees to the second axis. The reaction surface **310***a* may be angled towards the forward direction of the sliding latch **310**. The camming surface **330***a* and reaction surface **310***a* may be in contact in the first state where the button **331** of the first actuator **330** is not pressed. The resiliently deformable element **320** may be in an expanded state in the first state where the button **331** of the first actuator **330** is not pressed.

[0071] The personal hygiene tool **1** may include a blocker **340** which is engageable with the sliding latch **310** in the extended position to prevent the sliding latch **310** from moving towards the retracted position. The sliding latch **310** may include a blocking surface **310***b* which is in contact with the blocker **340** when the sliding latch **310** is in the extended position. The blocking surface **310***b* may face towards the rear end of the sliding latch **310** and may be prevented from moving in the rearwards direction by the blocker **340**. As shown, the blocker **340** may be arranged in a central recess of the sliding latch **310** and the blocking surface **310***b* may be an inner surface of the recess. In some examples, the blocking surface **310***b* may be formed at the rear end of the sliding latch **310**

or formed in an external recess e.g. formed in one side of the sliding latch **310**.
[0072] FIG. **6** is a longitudinal section schematic of the first end of the personal hygiene tool **1**.
[0073] The personal hygiene tool **1** may include a second actuator **350** configured to move the blocker **340** from an engaged position where the blocker **340** is engaged with the sliding latch **310** to a disengaged position where the blocker **340** is disengaged from the sliding latch **310**. The blocker **340** may be formed as a part of the second actuator **350**. In some examples, the second actuator **350** may be a separate component arranged to engage with and move the blocker **340**.
[0074] The second actuator **350** may include a user-depressible button **351** movable along the second axis, where an activation direction of each actuator is along the second axis towards the other actuator. The second actuator **350** may be supported by the lower latch support **212**. As shown, the user may apply an upwards force to move the button **351** and the second actuator **350** upwards along the second axis. FIG. **6** shows a second state where the button **351** of the second actuator **350** is pressed and the button **331** of the first actuator **330** is not pressed.
[0075] The button **351** of the second actuator **350** may be moved a distance of 0.8 mm inwards. In some examples, the distance moved by the button **351** of the second actuator **350** may be between 0.1 mm and 5 mm. The distance moved by the button **351** of the second actuator **350** may be greater or lesser than this range, as required.
[0076] As shown, the blocker **340** may be moved out of the engaged position by the second actuator **350**. The blocker **340** may be moved upwards from the engaged position to the disengaged position. An upper side of the sliding latch **310** may include a recess **311** aligned with the blocker **340** when the blocker **340** is in the disengaged position such that the sliding latch **310** can move in the rearward direction without making contact with the blocker **340**. In some examples, the blocker **340** in the disengaged position may be moved beyond the upper side of the sliding latch **310**.
[0077] The blocker **340** may be moved upwards towards the first actuator **330**. The first actuator **330** may have a recess to allow movement of blocker **340** in an upwards direction without making contact with the first actuator **330**. By providing the arrangement of the blocker **340** and the second actuator **350**, the tool head attachment **200** can be rotated by simultaneous actuation of the first actuator button **331** and the second actuator button **351**. This can prevent accidental unlatching leading to unintended rotation of the tool head attachment **200**. The arrangement of the first actuator **330** and second actuator **350** allows the respective actuator buttons **331**,**351** to be squeezed by the user e.g. by a finger and thumb pinch.
[0078] FIG. **7** is a longitudinal section schematic of the first end of the personal hygiene tool **1**.
[0079] The camming surface **330***a* may urge the sliding latch **310** in a direction perpendicular to the movement direction of the first actuator **330**. The sliding latch **310** may be moved in a rearward direction when the blocker **340** is in the disengaged position. FIG. **7** shows a third state where the button **351** of the second actuator **350** is pressed and the button **331** of the first actuator **330** is also pressed.
[0080] The button **331** of the first actuator **330** may be moved a distance of 0.8 mm inwards. In some examples, the distance moved by the button **331** of the first actuator **330** may be between 0.1 mm and 5 mm. The distance moved by the button **331** of the first actuator **330** may be greater than this range to provide greater security or less than this range to improve usability or user convenience.
[0081] The first actuator **330** may be moved downwards towards the second actuator **350**. The recess of the first actuator **330** may allow movement of the first actuator **330** in a downwards direction without making contact with the blocker **340** of the second actuator **350**.
[0082] The resiliently deformable element **320** may be in a partially or fully compressed state in the third state where the button **331** of the first actuator **330** is pressed. The sliding latch **310** may be moved a distance of 0.5 mm out of the recess **102** in the third state. In some examples, the distance moved by the sliding latch **310** may be between 0.1 mm and 5 mm. The distance moved by the button **351** of the second actuator **350** may be greater than this range to provide greater security or

less than this range to improve usability or user convenience. In some examples, the sliding latch **310** may be urged to the retracted position where the sliding latch **310** is disengaged from the recesses **102**.

[0083] In some examples, the first actuator **330** may be configured to urge the sliding latch **310** to a point between the extended position and the retracted position where the sliding latch **310** is partially engaged with one of the plurality of recesses **102**. In this position, the sliding latch **310** may be moved between recesses **102** under a rotational force applied by the user, and the user can receive a tactile sensation and/or an audible click when the sliding latch **310** moves into each recess **102**.

[0084] An outer edge of each recess **102** and a protruding end of the sliding latch **310** may be bevelled or rounded. An extent of the interference between the sliding latch **310** and the recess **102** in the third state may be 0.2 mm. The sliding latch **310** may be moved a further 0.2 mm under a rotational force applied by the user to move between recesses **102**. In some examples, the extent of the interference may be between 0.1 mm and 2 mm. The extent of the interference may be greater than this range to provide greater security or less than this range to improve usability or user convenience.

[0085] In another embodiment, the first actuator may be formed in a single piece with the upper latch support. The second actuator may be formed in a single piece with the lower latch support.

[0086] FIG. **8** is a perspective view of a first actuator **430** according to a second embodiment. Components of the second embodiment not described in detail may be substantially the same as the first embodiment. The first actuator **430** may include a protrusion **432** for connecting a user-depressible button. The user may apply a downwards force to move the button and the first actuator **430** downwards along the second axis.

[0087] The first actuator **430** may include a camming surface **430***a* set at an angle to a movement direction of the first actuator **430**. The camming surface **430***a* may be set at 45 degrees to the second axis. The camming surface **430***a* may be arranged to engage with the sliding latch **410**.

[0088] The first actuator **430** may include one or more interlocking elements **433**. The interlocking elements **433** may be arranged to interlock with elements of a second actuator **450**, when the first actuator **430** is adjacent to the second actuator **450**.

[0089] FIG. **9** is a longitudinal section schematic of the first end of the personal hygiene tool **1′**, according to the second embodiment. As shown, the personal hygiene tool **1′** may include the first actuator **430**, the second actuator **450** and a sliding latch **410**.

[0090] The sliding latch **410** may have a reaction surface **410***a* set at an angle to the second axis. The reaction surface **410***a* may be set at 45 degrees to the second axis. The reaction surface **410***a* may be angled towards the forward direction of the sliding latch **410**. The camming surface **430***a* and reaction surface **410***a* may be in contact in the first state where the button of the first actuator **430** is not pressed.

[0091] FIG. **10** is a perspective view of the second actuator **450** according to the second embodiment. The second actuator **450** may include a protrusion **452** for a user-depressible button. The user may apply an upwards force to move the button and the second actuator **450** upwards along the second axis.

[0092] The second actuator **450** may be configured to move a blocker **440** from an engaged position where the blocker **440** is engaged with the sliding latch **410** to a disengaged position where the blocker **440** is disengaged from the sliding latch **410**. The blocker **440** may be formed as a part of the second actuator **450**.

[0093] The second actuator **450** may include one or more interlocking elements **453**. The interlocking elements **453** may be arranged to interlock with interlocking elements **433** of the first actuator **430**, when the first actuator **430** is adjacent to the second actuator **450**.

[0094] FIG. **11** is a perspective view of the first end of the personal hygiene tool **1′**, according to the second embodiment.

[0095] The blocker **440** may be engageable with the sliding latch **410** in the extended position to prevent the sliding latch **410** from moving towards the retracted position. The sliding latch **410** may include a blocking surface **410***b* which is in contact with the blocker **440** when the sliding latch **410** is in the extended position. The blocking surface **410***b* may face towards the rear end of the sliding latch **410** and may be prevented from moving in the rearwards direction by the blocker **440**. As shown, the blocker **440** may be formed in an external recess e.g. formed in one side of the sliding latch **410**.
[0096] As shown, the blocker **440** may be moved out of the engaged position by the second actuator **450**. The blocker **440** may be moved upwards from the engaged position to the disengaged position. The sliding latch **410** may include a recess **411** aligned with the blocker **440** when the blocker **440** is in the disengaged position such that the sliding latch **410** can move in the rearward direction without making contact with the blocker **440**.
[0097] By providing the arrangement of the blocker **440** and the second actuator **450**, the tool head attachment can be rotated by simultaneous actuation of the first actuator button **432** and the second actuator button **452**. This can prevent accidental unlatching leading to unintended rotation of the tool head attachment. The arrangement of the first actuator **430** and second actuator **450** allows the respective actuator buttons **432**,**452** to be squeezed by the user e.g. by a finger and thumb pinch.
[0098] FIG. **12** is a partially disassembled view of a tool head attachment **200** of the personal hygiene tool **1**′ according to the second embodiment. As shown, the tool head attachment **200** includes the sliding latch **410**, the resiliently deformable element **320**, the second actuator **450** and a button **451** of the second actuator **450**.
[0099] The sliding latch **410** may be supported by the second actuator **450**, or the lower latch support **212** in the first embodiment. The first actuator **430** or the upper latch support **211** may be arranged to fit over the sliding latch **410** and attach to the second actuator.
[0100] The protruding end of the sliding latch **410** may have a circular form. The plurality of recesses **102** in the housing may have a corresponding circular form. The resiliently deformable element **320** may be arranged at the rear end of the sliding latch **410**.
[0101] The second actuator **450** may be fixed to rotate with the sliding latch **410**. The first actuator **430** may be fixed to rotate with the sliding latch **410**.
[0102] FIG. **13** is an oblique section schematic of the tool head attachment **200**′ of the personal hygiene tool **1**′ according to the second embodiment. As shown, the tool head attachment **200**′ includes the button **451** of the second actuator **450**, the second actuator **450**, the sliding latch **410**, the first actuator **430** and the button **431** of the first actuator **430**.
[0103] The personal hygiene tool **1**′ may include a second resiliently deformable element **460** arranged between the first actuator **430** and the second actuator **450** to resist movement in the respective activation direction of each actuator. In this way, the arrangement of the first and second actuators **430**,**450** can be constructed without an intermediate portion of the attachment frame **210** fixed between the actuators **430**,**450**, and with only a single resiliently deformable element **460**. This can simplify construction and reduce the number of components required.
[0104] The buttons **431**,**451** may be fixed to rotate with the respective actuators **430**,**450**. In some examples, the buttons **431**,**451** may be free to rotate with respect to the respective actuators **430**,**450**. In this way, the user's first hand can remain in position while the other hand rotates the attachment frame **210**.
[0105] One or more magnets **470** may be arranged within the button **431** of the first actuator **430** or the button **451** of the second actuator **450**. The magnets **470** may allow the personal hygiene tool **1**′ to be fixed in position e.g. in a case.
[0106] FIG. **14** is a lateral section schematic of the first end of the personal hygiene tool **1**.
[0107] As shown, the sliding latch **310** may be urged to a point between the extended position and the retracted position where the sliding latch **310** is partially engaged with one of the plurality of recesses **102**.

[0108] The plurality of recesses **102** may be arranged in an arc extending in a plane perpendicular the second axis. The handle **100** may include five recesses **102** spaced equally apart. In some examples, the number of recesses **102** may be greater or less than five.
[0109] The tool head attachment **200** may define a third axis extending between the second axis and a working point of the tool. The sliding latch **310** may be arranged to extend at 45 degrees to the third axis. The arrangement can allow the tool head attachment **200** to be rotated from a starting position aligned with the first axis to a right-angled position, while the recesses **102** are equally spaced around the first axis.
[0110] FIG. **15** is a partially disassembled view of a blade holder **10** of the personal hygiene tool **1**. The blade holder **10** may include a frame **11**, one or more blades **12** and a cover **13**.
[0111] The frame **11** may include an engaging element configured to engage with a corresponding receiving portion on the attachment frame **210** of the tool head attachment **200**. The engaging element may include a depressible element which can be depressed as the blade holder **10** is engaged with the tool head attachment **200** and engages with the receiving portion of the attachment frame **210** to resist removal of the blade holder **10**. The depressible element may be arranged to be pressed by the user to disengage from the receiving portion and remove the blade holder **10** from the tool head attachment **200**.
[0112] The frame **11** may include a recessed area shaped to receive the one or more blades **12**. The recessed area may include one or more protrusions configured to engage with the blades **12**. The one or more blades **12** may be formed with one or more through holes configured to receive the protrusions of the frame. The blades may be formed from stainless steel or any other suitable material. The cover **13** may be configured to attach to the frame **11** with the blades **12** between. The cover **13** may be configured to attach to the protrusions of the frame **11** to fix the blades **12** in position.
[0113] FIG. **16** is a perspective view of the personal hygiene tool **1** with a charging base **2**. The handle **100** may be formed from metal or plastic materials. In some examples, the front handle portion **110**, rear handle portion **120** and end cap **140** may be formed from stainless steel or anodised aluminium. The outer grip **130** may be formed from rubber, rubberised or rubber-coated plastic.
[0114] The tool head attachment **200** may be formed from plastic or machined metal components e.g. machined aluminium.
[0115] The charging base **2** is formed to support the personal hygiene tool **1** thereon and provide power to the personal hygiene tool **1**. The charging base **2** may be configured to support the handle **100** of the personal hygiene tool **1** in a vertical orientation. The charging base **2** may include an opening **21** configured to receive the rear handle portion **120** of the personal hygiene tool **1**. The charging base **2** may be configured to charge the personal hygiene tool **1** when the personal hygiene tool **1** is supported by the charging base **2**.
[0116] FIG. **17** is a disassembled view of the charging base **2**. The charging base **2** may include a housing **20**, charging assembly **30**, bottom cover **40** and friction ring **50**.
[0117] The housing **20** may be formed with an upper cavity **21** and a lower cavity **22**. The upper cavity **21** may be open to an upper side of the housing **20** and may be configured to receive the rear handle portion **120** of the personal hygiene tool **1**. The upper cavity **21** may be shaped to correspond with the rear handle portion **120** and end cap **140** of the handle **100**. For example, the upper cavity **21** may be formed with a sloped floor to abut with an angled end cap **140** of the handle **100**. The upper cavity **21** may be formed with a drainage hole through to the lower cavity **22**.
[0118] The lower cavity **22** may be open to a lower side of the housing **20** and may be configured to receive the charging assembly **30**. The charging assembly **30** may include a wireless charging coil **31**, a PCB **32**, and a charging port **33**. The wireless charging coil **31** may be arranged adjacent to the upper cavity **21** to interface with a corresponding charging coil in the handle **100** of the

personal hygiene tool **1**. The PCB **32** may be connected to the wireless charging coil **31** and charging port **33** and may include a controller to manage charging of the personal hygiene tool **1**. The charging port **33** may be a USB port e.g. a USB-C port. The housing **20** may include a side opening aligned with the charging port **33** for access thereto.

[0119] The bottom cover **40** may be arranged to cover the lower cavity **22**. The bottom cover **40** may be formed with a drainage hole aligned with the drainage hole of the upper cavity **21**. The bottom cover **40** may include a ring-shaped groove to retain the friction ring **50**. The friction ring **50** may be formed of rubber, rubberised plastic or silicone and may be arranged on the bottom cover **40** to prevent the charging base **2** from slipping.

[0120] FIG. **18** is a plan view of the personal hygiene tool **1** with a charging case **3**. The charging case **3** is formed to hold the personal hygiene tool **1** inside and provide power to the personal hygiene tool **1**. The charging case **3** may be configured to support the personal hygiene tool **1** in a horizontal orientation. The charging case **3** may include a charging groove **71** configured to receive the rear handle portion **120** of the personal hygiene tool **1**. The charging case **3** may be configured to charge the personal hygiene tool **1** when the personal hygiene tool **1** is supported within the charging case **3**. The charging case **3** may be configured to sterilise the blade holder **10** of the personal hygiene tool **1** when the personal hygiene tool **1** is supported within the charging case **3**.

[0121] FIG. **19** is a disassembled view of the charging case **3**. The charging case **3** may include a housing **60**, an inner shell **70**, a charging assembly **80** and a sterilising assembly **90**. The housing **60** may be formed from an upper shell **61** and a lower shell **62**. The upper shell **61** and lower shell **62** may be attached with a hinge along one side to form a clamshell. One or both of the upper shell **61** and lower shell **62** may include one or more magnets to keep the housing **60** closed. The upper shell **61** and lower shell **62** may be formed from plastic. An outer surface of the upper shell **61** and lower shell **62** may be rubberised or rubber coated.

[0122] The inner shell **70** may be formed to support the personal hygiene tool **1** within the housing **60**.

[0123] The inner shell **70** may include a charging groove **71** configured to receive the rear handle portion **120** of the personal hygiene tool **1**. The inner shell **71** may also include a retaining groove **72** configured to position the handle **100** of the personal hygiene tool **1**. One or more positioning magnets may be arranged within or beneath the inner shell **70** to hold the handle **100** in position.

[0124] The inner shell **70** may include a blade holding area **73** formed to hold one or more additional blade cartridges. The inner shell **70** include an area for the charging assembly **80** and an area for the sterilising assembly **90**.

[0125] The charging assembly **80** may include a wireless charging coil **81**, a PCB **82**, a charging port **83**, and a battery **84**. The wireless charging coil **81** may be arranged adjacent to the charging groove **71** to interface with a corresponding charging coil in the handle **100** of the personal hygiene tool **1**. The PCB **82** may be connected to the wireless charging coil **81** and charging port **83** and may include a controller to manage charging of the personal hygiene tool **1**.

[0126] The charging port **83** may be a USB port e.g. a USB-C port. The lower shell **62** may include a side opening aligned with the charging port **83** for access thereto. If a power source is connected to the charging port **83**, the controller of the PCB **82** may be configured to charge the personal hygiene tool **1** and charge the battery **85** when the personal hygiene tool **1** is fully charged. If a power source is not connected to the charging port **83**, the controller of the PCB **82** may be configured to charge the personal hygiene tool **1** using the battery **85**.

[0127] The sterilising assembly **90** may include one or more sterilising LEDs **91** and a Hall sensor **92**. The LEDs **91** may be ultraviolet (UV) LEDs, e.g. UV-C LEDs, arranged near the blade holder **10** of the personal hygiene tool **1** when the handle **100** is correctly positioned in the housing **60**.

[0128] The Hall sensor **92** may be mounted on the PCB **82** or elsewhere on the inner shell **70** and may be aligned with a magnet in the upper shell **61**. In this way, the Hall sensor **92** can detect whether the housing **60** is open or closed. The controller of the PCB **82** may be configured to activate the LEDs **91** when the housing **60** is closed and deactivate the LEDs when the housing **60** is open.

[0129] FIG. **20** is a perspective view of a personal hygiene system **1000** including the personal hygiene tool **1**, as well as the charging base **2** and/or charging case **3**.

[0130] The above embodiments are described by way of example only. Many variations are possible without departing from the scope of the disclosure as defined in the appended claims.

[0131] For clarity of explanation, in some instances the present technology may be presented as including individual functional blocks including functional blocks comprising devices, device components, steps or routines in a method embodied in software, or combinations of hardware and software.

[0132] Methods according to the above-described examples can be implemented using computer-executable instructions that are stored or otherwise available from computer readable media. Such instructions can comprise, for example, instructions and data which cause or otherwise configure a general purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. Portions of computer resources used can be accessible over a network. The computer executable instructions may be, for example, binaries, intermediate format instructions such as assembly language, firmware, or source code. Examples of computer-readable media that may be used to store instructions, information used, and/or information created during methods according to described examples include magnetic or optical disks, flash memory, Universal Serial Bus (USB) devices provided with non-volatile memory, networked storage devices, and so on.

[0133] Devices implementing methods according to these disclosures can comprise hardware, firmware and/or software, and can take any of a variety of form factors. Typical examples of such form factors include laptops, smart phones, small form factor personal computers, personal digital assistants, and so on. Functionality described herein also can be embodied in peripherals or add-in cards. Such functionality can also be implemented on a circuit board among different chips or different processes executing in a single device, by way of further example.

[0134] The instructions, media for conveying such instructions, computing resources for executing them, and other structures for supporting such computing resources are means for providing the functions described in these disclosures.

[0135] Although a variety of examples and other information was used to explain aspects within the scope of the appended claims, no limitation of the claims should be implied based on particular features or arrangements in such examples, as one of ordinary skill would be able to use these examples to derive a wide variety of implementations. Further and although some subject matter may have been described in language specific to examples of structural features and/or method steps, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to these described features or acts. For example, such functionality can be distributed differently or performed in components other than those identified herein. Rather, the described features and steps are disclosed as examples of components of systems and methods within the scope of the appended claims.

Claims

1. A personal hygiene tool comprising: a handle extending along a first axis and comprising an opening formed at a first end, where a plurality of recesses are formed within the opening; and a tool head attachment at least partially retained within the opening and rotatable with respect to the handle around a second axis; wherein the tool head attachment comprises a latching mechanism comprising: a sliding latch slideable between an extended position where the sliding latch protrudes into one of the plurality of recesses and a retracted position where the sliding latch is disengaged from the recesses; a resiliently deformable element arranged to urge the sliding latch towards the extended position; and a first actuator configured to urge the sliding latch towards the retracted position against the resiliently deformable element.

2. The personal hygiene tool of claim 1, wherein the first actuator comprises a user-depressible button movable along the second axis.

3. The personal hygiene tool of claim 1, wherein the first actuator comprises a camming surface set at an angle to a movement direction of the first actuator, and the camming surface is arranged to engage with the sliding latch and urge the sliding latch in a direction perpendicular to the movement direction of the first actuator.

4. The personal hygiene tool of claim 1, wherein the sliding latch is an elongate element slideable along its length in a plane perpendicular the second axis.

5. The personal hygiene tool of claim 1, wherein an outer edge of each recess and a protruding end of the sliding latch are bevelled or rounded, and wherein the first actuator is configured to urge the sliding latch to a point between the extended position and the retracted position where the sliding latch is partially engaged with one of the plurality of recesses.

6. The personal hygiene tool of claim 1, wherein the plurality of recesses and a protruding end of the sliding latch have a circular form.

7. The personal hygiene tool of claim 1, further comprising a blocker which is engageable with the sliding latch in the extended position to prevent the sliding latch from moving towards the retracted position.

8. The personal hygiene tool of claim 7, further comprising a second actuator configured to move the blocker from an engaged position where the blocker is engaged with the sliding latch to a disengaged position where the blocker is disengaged from the sliding latch.

9. The personal hygiene tool of claim 8, wherein the second actuator comprises a user-depressible button movable along the second axis, where an activation direction of each actuator is along the second axis towards the other actuator.

10. The personal hygiene tool of claim 9, further comprising an second resiliently deformable element arranged between the first actuator and the second actuator to resist movement in the respective activation direction of each actuator.

11. The personal hygiene tool of claim 1, wherein the plurality of recesses are arranged in an arc extending in a plane perpendicular the second axis.

12. The personal hygiene tool of claim 1, wherein the tool head attachment defines a third axis extending between the second axis and a working point of the tool, and the sliding latch is arranged to extend at 45 degrees to the third axis.

13. The personal hygiene tool of claim 1, wherein the second axis is perpendicular to the first axis.

14. The personal hygiene tool of claim 1, wherein the tool is a shaver, trimmer or scraper, and the tool head attachment comprises a blade or blade assembly.

15. A personal hygiene system comprising the personal hygiene tool of claim 1, and at least one of: a charging base formed to support the personal hygiene tool thereon and provide power to the personal hygiene tool; and a charging case formed to hold the personal hygiene tool inside and provide power to the personal hygiene tool.